Northeastern University, Khoury College of Computer Science

CS 6120 — Some Extra Practice 5

Due: March 23, 2026(100 points)

Practice Quiz 5 - Advanced Sequence Modeling

Instructions: Closed book. No calculators. Show all work. All probabilities are given in standard (not log) space unless stated otherwise.

Consider a simplified linear recurrent neural network: $h_k = Wh_{k-1}$ where $h_k \in \mathbb{R}^d$ and $W \in \mathbb{R}^{d\times d}$. Let's say that the true parameter W matrix can be represented as:

$$W = \begin{pmatrix} 2.0 & 0 \\ 0 & 0.5 \end{pmatrix} \tag{1}$$

Problem 1: Matrix Properties

With the W defined in (1) where $W_{ij} = 0$ for $i \neq j$, derive an expression for the $(i,j)^{\text{th}}$ component of the gradient $\frac{\partial h_K^{(i)}}{\partial h_0^{(j)}}$ in terms of K.

Problem 2: Directional Behavior

Describe what happens to the vector gradient $\frac{\partial h_K}{\partial h_0}$ as $K \to \infty$.

Problem 3: The Identity Case

Suppose $W = I$ (the identity matrix).

1. What is the value of λ in this case?
2. Provide the exact expression for h_K in terms of h_0.

3. Does this system exhibit vanishing or exploding gradients?

Problem 4: Scalar Simplification

Assume $d = 1$ (so W, h are scalars). Let $h_k = wh_{k-1}$ and $w = 0.5$.

1. Calculate the ratio $\frac{h_{10}}{h_0}$.
2. If we want to ensure h_K does not drop below 1% of h_0, what is the maximum allowable value for K?

Solutions

Problem 1 Solutions

In a linear RNN, $h_T = W^T h_0$. For a diagonal matrix W, this separates into $h_T^{(i)} = \lambda_i^T h_0^{(i)}$. The partial derivative is: $\frac{\partial \mathbf{h_T^{(i)}}}{\partial \mathbf{h_0^{(i)}}} = \lambda_\mathbf{i}^\mathbf{T}$.

Problem 2 Behavior:

As $T \to \infty$:

- In the first dimension ($i = 1$), the gradient component $\mathbf{2.0^T}$ **explodes**.
- In the second dimension ($i = 2$), the gradient component $\mathbf{0.5^T}$ **vanishes**.

The gradient vector norm explodes overall, but the vector itself becomes increasingly aligned with the first axis as the second component disappears.

Problem 3 Solutions

(1) $\lambda = 1$. (2) $h_K = h_0$. (3) Neither; the gradient remains constant.

Problem 4 Solutions

(1) $0.5^{10} \approx 0.00097$. (2) $0.5^K \geq 0.01 \implies K \leq \frac{\log(0.01)}{\log(0.5)} \approx 6.64$, so $K_{max} = 6$.